Differentiating Accumulation Functions with the FTC

Directions: An accumulation function is a function defined by an integral with a variable limit, e.g. $g(x) = \int_a^x f(t)\,dt$. The Fundamental Theorem says $g'(x) = f(x)$: the integrand *is* the derivative. Show the step that names which form of the theorem you are using, then compute. Give exact values unless a decimal is asked for; state units where the problem gives them.

This workshop covers: Differentiating accumulation functions, evaluating accumulated change, the chain-rule form of the FTC, and reading an accumulation function's behaviour.

1. Let $g(x) = \int_1^x \left(t^3 + 2\right) dt$.

Find $g'(3)$. (You should not need to evaluate the integral.)

Answer: ________________

2. Water flows into a tank at the rate $r(t) = 6t + 4$ litres per minute, where t is measured in minutes. The volume added during the first x minutes is $V(x) = \int_0^x (6t + 4)\,dt$.

Find $V(5)$, the volume added during the first 5 minutes.

Answer: ________________ L

3. Let $g(x) = \int_1^{x^2} \sqrt{1+t^3}\,dt$.

Find $g'(2)$. Give the exact value.

Answer: ________________

4. A continuous, decreasing function f has the values in the table, and $g(x) = \int_0^x f(t)\,dt$ for $0 \le x \le 7$.

t	0	1	2	3	4	5	6	7
$f(t)$	4	3	2	0	-2	-3	-4	-5

On which intervals is g increasing, and on which is it decreasing? Where on $[0, 7]$ does g attain its absolute maximum? Justify each claim with the Fundamental Theorem — this one is graded on the reasoning, not just the answer.

5. Let $h(x) = \int_x^4 \ln\left(t^2 + 1\right) dt$. Note that the variable is the **lower** limit.

Find $h'(2)$. Give the exact value.

Answer: ____________

6. A pump adds water to a reservoir at the rate $r(t) = 12 - t$ litres per minute for $0 \le t \le 10$, and $V(x) = \int_0^x (12 - t)\, dt$ is the volume added in the first x minutes.

(a) Find $V(10)$, in litres.

Answer: ____________

(b) Find $V'(10)$, and state what it tells you about the reservoir at $t = 10$ minutes, with its units.

Answer: ____________

7. Let $g(x) = \displaystyle\int_0^x \left(4t^3 - 6t\right) dt$.

(a) Evaluate the integral to write an explicit formula for $g(x)$ (a polynomial in x, no integral sign).

Answer: ________________

(b) Differentiate your formula from part (a) to find $g'(x)$, and say what the Fundamental Theorem predicted the answer would be.

Answer: ________________

8. **Synthesis.** Let $g(x) = \displaystyle\int_1^{x^2} \frac{3}{t}\, dt$ for $x > 0$.

(a) Find $g'(x)$ and simplify it completely.

Answer: ________________

(b) Evaluate $g'(2)$.

Answer: ________________